



EMPOWERING ZAMBIA'S SMALL SCALE FARMERS THROUGH MARKET TRANSPARENCY AND CONNECTIVITY

Team techdex

1. MERCY MAINA
2. FATMA MOHAMED
3. MARIA NYANUNGO
4. DIVINE IGIHOZO



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1. INTRODUCTION

While Zambia's global identity is often tied to its mineral wealth, agriculture serves as the nation's true socioeconomic heartbeat. The sector provides livelihoods for approximately 70% of the population and supports over 1.6 million small-scale farming households, who collectively produce nearly 80% of the country's food. However, a stark contradiction exists: as noted by the Zambia Development Agency (2024), the very producers responsible for national food security remain the most economically vulnerable due to a lack of basic digital tools and equitable market access.

Zambia was an early pioneer in the digital frontier, launching the Zap mobile payment system in 2002 predating Kenya's M-Pesa by half a decade. Yet, despite this head start, the last mile connectivity to remote rural villages remains unfulfilled. According to the World Bank (2025), while urban centers experience rapid digitalization, smallholders are increasingly marginalized by a combination of climate change and technological isolation. This digital divide prevents farmers from accessing real-time market data, leaving them unable to negotiate fair prices and forcing them into a cycle of survival mode. With less than 0.5% of smallholders having access to formal credit (IFAD, 2024) and high infrastructure costs limiting adoption (ZICTA, 2024), digital transformation is no longer a luxury; it is a survival imperative for Zambian agriculture.

The primary purpose of this report is to investigate the symmetry barriers that prevent zambian small holder farmers from achieving economic stability with a focus on the role of digital exclusion. By analyzing the intersection of technology, infrastructure ,and market dynamics, this report aims to provide a clear understanding of why small scale farmers in Zambia are remaining economically marginalized despite their role in Zambia GDP

To achieve this, The report focuses on finding out how missing market information affects the farmers prices, looking at how lack of digital knowledge keeps farmers hidden from the big buyers, and seeing how solutions like the Maano app are actually helping. Overall , the report shows how to give power back to the farmers.

PROBLEM ANALYSIS

The core problem is Market Accessibility. While farmers produce 80% of Zambia's food, they are physically and digitally locked out of the markets where they could earn a fair price.

PROBLEM IDENTIFICATION

The central issue is the exclusion of small holders farmers from the high value trading cycle. This is not about not having somewhere to sell, but the lack of connection. Farmers are not able to reach and talk to the big market but they are stuck to the surviving markets that force them to deal with middlemen who take advantage of the farmer's invisibility and keep the prices low.

ANALYSIS

The Infrastructure Barrier

Farmers may see high value opportunities on an app, but they remain an average of 30.2 km away from a tarred road (Mercy Corps Agrifin, 2020). The market remains unreachable because of the transport costs

The Unreliable information

Market isolation strips farmers of their bargaining power. Without a reliable way to reach multiple buyers or store their goods safely. Even with introduction to mobile trading platforms, the absence of real time data on alternative buyers means farmers cannot wait for a fair offer. Zambia Development (2024), This lack of choice creates a cycle of poverty .

LITERATURE REVIEW

The primary barrier to small holder prosperity is information asymmetry (IA). This imbalance creates a shroud of invisibility, where critical data held by middlemen is withheld from farmers,

preventing them from participating in the economy as equal partners. This structural breakdown manifests in three critical ways:

1. The unreliable Information and pricing Transparency

Without a centralized , real time database of national market rates, smallholders rely on localization unreliable information to value their crops. Thai lack of transparency allows buyers to exploit the farm gate gap, offering prices that fail to compensate farmers for their labor and inputs

2. Invisibility gap

Farmers cannot see high-value markets, and big commercial buyers or NGOs cannot see the farmers. Because smallholders lack digital records regarding harvest cycles and crop quality. This exclusion keeps even high quality products out of the formal value chain.

3. Bargaining power

Information is the cornerstone of negotiation. When a farmer is unaware of alternative buyers, they lose their status as a businessman and become a price taker. This desperation forces them to sell to the first person who pays them, perpetuating a vicious cycle of poverty

2. THE FINDINGS

Agriculture in Zambia accounts for 20% of the GDP with 12% being national export earnings. This sector employs 70% of the Zambian labour force and is the primary source of

income for most families in Zambia(FtMa, 2020). Despite this they face various challenges that will be discussed.

Small Holder farmers in Zambia, approximately 1.6 million of Zambians, 20% being women , are hindered by complex structural,physical and informational challenges that push them to the margins of the national economy. The primary driver to this marginalisation is the information disconnect where important market data available to traders and middlemen is made unavailable to the small holder farmers.This difference shrouds the farmers in a cloak of invisibility preventing them from participating as equal contributors to the economy of Zambia. This damages the pricing transparency(Imani Development, 2025)

One out of six small holder farmers own a mobile phone most lack the connectivity to find alternative buyers(Mercy Corps AgriFin, 2020). Instead of having access to real time data on the national market prices, farmers have to rely on community rumors to determine crop values, this creates an opportunity for middlemen and buyers to offer “farm gate” prices that do not fairly compensate the farmers for their labour(Imani Development, 2025)The farmers have a heavy reliance on informal markets where prices are low risks are high and opportunity for growth is severely restricted.

This invisibility creates a two way barrier between farmers and organisations and NGOs. While the farmers cannot see the broader market, larger commercial buyers and NGOs often view farmers as untraceable and risky because of a lack of digital data on the crop health and quality and their harvest schedules.(Imani Development, 2025). The lack of any digital data related to a farmer's crop, when it is expected to be harvested, how the crop is expected to be evaluated is a reason why the high-paying organizations are seen as high-risk and "untraceable" when it comes to smallholders. Small farmers, even if their products are of good quality, are unable to get into the formal value chain due to both the blindnesses.Even when farmers produce high value goods, they are often unable to enter the formal value chains since they lack technical skills and mandatory sanitary and phytosanitary regulations like GlobalGAP(Akiri et al, 2024, as cited in Imani Development, 2025)

The fragmented nature of smallholder production makes it difficult for individual farmers to meet the large, consistent volumes required by formal processors and retailers (Abraham et al., 2022, as cited by Imani Development, 2025). Small holder farmers average just 1 tonne of soybeans per hectare, far below the 2.8 tonnes achieved by commercial farms (Imani Development, 2025). This is driven by lack of mechanization—only 0.2% of small holder farmers own a tractor—and an overstretched extension system with a ratio of 1 worker per 1000 farmers (Imani Development, 2025).

The resulting loss of bargaining power transforms the farmer from a business man into a “price taker” (Imani Development, 2025). Without knowledge of alternative buyers or higher market prices, farmers are often forced to sell their products to the first person they see that arrives with a truck and cash, perpetuating a vicious cycle of poverty (Imani Development, 2025). The most valuable assets in negotiating are confidence, trust and information. The results show that the “invisibility” directly affects the negotiation power. If a farmer doesn't know of a buyer who is willing to pay a higher price, then he or she must sell to the first person to come with a truck and cash. Lack of information on other markets turns the farmer from a businessman to a “price taker”, and he is always in a vicious circle of poverty.

Farmers live an average of 12km from the nearest financial agent and must travel 30.2km to reach a tarred road (Mercy Corps AgriFin, 2020; Imani Development, 2025). These gaps contribute to the post harvest losses that they face very harvesting season that reach up to 50% in some households. (Imani Development, 2025) Approximately 75% of small holder farmers are vulnerable to extremes in the climate as evidenced in the recent flood that occurred in Chitamo which destroyed 40 hectares of crops, impacting the food and income of 70 households (Mwila & Mbewe, 2022; Imani Development, 2025).

Failure of Current Information Delivery – failure to deliver the information needed by the user when needed. Existing programmes to bridge this gap are failing to do so because they do not offer access or are not in an appropriate form. Government agencies often don't publish the market data in a way that can be easily consumed, it's difficult to find and listen to the radio at the wrong time, and it's typically costly to be able to obtain it on a smartphone. The majority of

the 16 million people involved in the farming process are not genuinely addressing the essence of the asymmetry problem and the "information bridge" is not going to be accessible to the most vulnerable farmers.

The Enterprise Zambia Challenge Fund has enabled 284,117 farmers to access markets while the ZIAMIS platform now covers 4.3 million verified and traceable farmers (FAO, 2025; Imani Development, 2025). This is a significant number but it does not mean 100% coverage meaning that more than 75% of the small holder farmers are not accounted for. The Maano Virtual Farmers market has also expanded to 99,016 farmers by 2023. They have also ensured that farmers receive digital advisories via ZIAMIS. This led to maize productivity rising to 3.4 tonnes per hectare which was above 2.14 national average (FAO, 2025)

While overall the financial inclusion has risen recently to 69.4% in 2020, traditional bank account ownerships remain at a below 10%, with most farmers opting to using mobile money which caused a surge of 58.4% mobile money penetration (Fintech News Africa, 2026). Despite all these digital tools information delivery often fails because it lacks user-friendliness and are often insufficient for in-person training (Mercy Corps AgriFin, 2020)

3. THE PROPOSED SOLUTION

The UN World food Programme together with the ZANACO launched an app called Maano with the aim of solving the problem of poverty in small scale farmers in Zambia. This app is a form of connection between the farmers and their customers who they might not be able to reach using the traditional method of selling. This app is built in such a way that the farmers are able to advertise their produce on the platform. This app is integrated with a knowledge system that provides awareness for customers so that they know how to price their produce. The app is available on playstore and can be accessed for download through a code *227# in Zambia (United Nations, 2021).

Zam Farm to Market is another app that was developed for Zambian rural farmers that also bridges the gap between customers and farmers. This platform provides a secure platform where farmers and buyers can transfer money without insecurities. The main goal of this initiative is to increase the market for rural farmers. Due to the problems with infrastructure, the Zam farm to market app provides a platform where farmers can connect with transportation. The app provides dispute support and governs how farmers are paid so as to avoid customers that do not pay. The farmers are verified to sell using the app to improve trust. The digital trade records are conserved on the app. This app has a standardisation which means it assesses the quality of the produce being sold. The app provides escrow style holding to increase integrity.

F-Hub is a platform designed for Zambian farmers that provide them with a commodity market. This platform uses a USSD and web platform to provide various services. Farmers are able to hire a tractor, take loans, convenient access to extension officers and the farmers get a chance to keep their records on this platform. Farmers register and they can access information on various aspects of farming. This app solves the problem of not accessing markets to buy quality services for customers (*Digital Platform – Farmer's Hub*, 2021).

Agriconnet Zambia is another platform that offers information and buying and selling facilities for small scale farmers in Zambia. This platform connects farmers so that they can share ideas and it is also a platform with job opportunities. This platform solves the problem of access to opportunities and markets for small scale farmers. All these platforms may vary in one way or the other but their main goal is to reduce the poverty levels in Zambian small scale farmers through access to a broader market. Some of these apps are adopted to a greater extent and some are not. This means there is a gap of trustworthiness among farmers and digital apps and platforms.

The proposed solution is an app and platform that will be built in such a way that it will accommodate farmers, non-governmental organisations and transport links that work with small scale farmers. The organisations are there to connect farmers with quality produce to the buyers who are willing to pay reputable amounts. The purpose of the organisations ensures that there is

reduction of conflicts such as farmers not receiving their payments. The purpose of the app is to create a direct connection between farmers and the organisations. The app also works as a platform to connect farmers with transportation such as Swift e.t.c. The solution becomes disruptive because there is no other app that connects farmers with the direct organisations. This improves farmers' trustworthiness towards digitalisation since they are working with prominent organisations.

4. RESEARCH METHODOLOGY

This report uses a qualitative secondary research approach. The study focuses on analyzing the existing literature, industry reports and case studies to understand the current state of digital adoption among small scale farmers in Zambia.

1. Data Sources

To ensure that the findings are current, data was collected from a variety of reputable source including:

International Organizations which includes reports from the world bank(2025),IFAD(2024),and United Nations World Food Programme(WFP).

Government and National Agencies like data from the Zambia Development Agency (ZAD) and the Zambia Information and Communications Technology Authority (ZICTA).

Industry Databases like Fintech market insights from Tracxn(2026) and Fintech News Africa.

Case Studies-specific analysis of digital platforms such as Maano ,Zanaco,Zoona and Lupiya.

2. Data Collection Criteria

The team searched for information using specific keywords such as "Zambia digital agriculture", "mobile literacy in rural Zambia", and "market access for Zambian farmers". We prioritized

documents published between 2020 and 2026 to capture both historical context (e.g the 2002 Zap launch) and future projections.

3. Data Analysis

The collected information was analyzed using thematic analysis. The data was categorized into four key themes:

- The gap between urban digital growth and rural agricultural stagnation.
- The effectiveness of digital marketplaces (e.g., Maano and FtMA) in connecting farmers to regional markets.
- Socio-economic barriers, specifically digital illiteracy and infrastructure costs

4. Limitations

As this is a desk-based review, the study is limited by the availability of online data. Direct field interviews with Zambian smallholders were not conducted; therefore, the report relies on the accuracy of the cited third-party researchers and agencies.

5. CRITICAL EVALUATION

1. Strategic Alignment of the proposed solution with the problem.

The proposed solution directly addresses the primary finding of information asymmetric by acting as a digital mediator. While the current system like Maano and Zam Farm to Market provide basic transactional infrastructure, the proposed solution introduces a unique layer of organizational verification. The strength of the solution is that by integrating the NGOs and verified organizations the app dismantles the invisibility issues for the small scale farmers as it transforms untraceable farmers into verified assets solving the problem where small scale farmers were previously viewed as risk because of lack of digital data. Unlike F-Hub which focuses on inputs like tractors and loans this solution focuses on the output value chain targeting the gap between the quality producers and reliable buyers.

2. Market Power and Farmers as Price Takers

The critical success factor of this proposed solution is the ability to shift the farmer from a price taker to an active market participant. The inclusion of escrow-style holdings that is similar to Zam Farm to Market and quality assessment by organizations address the findings that confidence and trust are most valuable assets in negotiation. The Disruptiveness of the proposal is the three way connection which involves farmers, NGOs and Transporters which is different from the traditional use of middlemen who currently benefit from misinformation. The proposed solution replaces unreliable information with a verified quality grading system.

3. USSD Adoption

The findings states that smartphone access is costly and government data is difficult to consume if the proposed app relies on smartphone features there is a risk of excluding the most vulnerable of the 16 million people in the sector therefore for it to be more effective the app must mirror the accessibility of the Maano USSD code (*227#).

4. Comparison with existing solutions

The proposed solution shifts from transactional to relational support by addressing the 95% of Zambian small scale holders who are often left behind by existing apps like Maano which while reducing the post-harvest losses, they relied on a pilot to serve 1000 to 5000 farmers. F-Hub focuses on productivity inputs, the new solution leverages professional NGO validation to bridge the trust gap as research shows networking and information hold greater success coefficients(0.2617)than technology only which has (0.1779).

5. Conclusion of Evaluation

The proposal is highly effective at addressing the qualitative aspects of the problem, specifically the lack of bargaining power and market visibility, however its success depends on the technical inclusivity. To ensure the app does not become a failed information bridge, the app must ensure its quality assessment and organization connection features are accessible via low bandwidth or USSD channels to reach the farmers currently trapped in a circle of poverty.

6.CONCLUSION

The struggle of Zambia's 1.6 million small -scale farmers is not because they are not working hard it is because they are disconnected. This report shows that even though these farmers grow most of the country's food ,they are trapped by lack of information, poor infrastructure and rely on unreliable information on the worth of their crops which makes them accept whatever low price a buyer offers as they don't know any better options.

The solution has suggested turning information into power by giving farmers real-time market prices and a way to prove the quality of their crops online making them becoming visible.

However the government and partners have to fix the physical gaps like the long distances farmers have to travel to find banks or paved roads.

In the end bringing small scale farmers into the digital world is the effective way to help Zambia grow fairly. By clearing away the fog of information we can make sure the people who feed the nation are finally able to earn a fair living and build a better future for their families.

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